

INDRA CHOUDHARY

Backend & AI/ML Engineer | MCA Candidate

Mysuru, Karnataka | +91-8669955641 | indrachoudhary362@gmail.com
linkedin.com/in/indrachoudhary29 | github.com/indrachoudhary29

EDUCATION

JSS Science and Technology University, Mysuru — MCA, 1st Yr (CGPA: 8.5/10)	2025 – 2027 (Expected)
St. Joseph's First Grade College, Mysuru — BCA (CGPA: 8.45/10)	2023 – 2025
Vivekananda P.U. College, Mysuru — PCMC (92.1%)	2020 – 2022

TECHNICAL SKILLS

Languages: Python, SQL, Bash

Backend & APIs: FastAPI, Flask, REST APIs, Celery, Redis

AI/ML: LangChain, RAG, Prompt Engineering, Vector Embeddings, Semantic Search, TensorFlow, OpenCV

Databases: MySQL, Qdrant (Vector DB)

Cloud & DevOps: Docker, AWS (EC2, S3 — basics), Git/GitHub, Linux

Core CS: Data Structures, Object-Oriented Programming, Operating Systems

PROJECTS

DocuSense AI — RAG-Based Document Assistant *Python, FastAPI, LangChain, Qdrant, MySQL, Celery, Redis, Docker*

- Architected a Retrieval-Augmented Generation pipeline that ingests PDF documents, chunks and embeds their content, and answers natural-language questions through a FastAPI backend exposing REST endpoints.
- Implemented semantic search over document embeddings using Qdrant as the vector store, enabling low-latency retrieval of relevant passages across large PDF collections.
- Orchestrated retrieval and generation with LangChain, routing queries to a Groq-hosted Llama 3.1 8B model and grounding responses in retrieved context with citations to reduce hallucinations.
- Offloaded document ingestion and embedding generation to asynchronous Celery workers backed by Redis, keeping the API responsive under concurrent uploads.
- Modeled user sessions and document metadata in MySQL, and built a Streamlit interface for end-to-end testing of the ingestion-to-answer pipeline.
- Designed the system for free-tier cloud deployment (Render/Railway, Qdrant Cloud), keeping infrastructure cost at zero.

Gender Identification through Ear Biometrics *Python, OpenCV, TensorFlow*

- Trained a convolutional deep learning model to classify gender from ear biometric images, reaching 85% accuracy on the test set.

Bag of Memories — Full-Stack Web Platform *Python, Flask, HTML, CSS, JavaScript*

- Built a full-stack memory-sharing web application with a Flask backend and an HTML/CSS/JavaScript frontend, handling user-submitted content and page rendering end-to-end.

Data Automation Script *Python*

- Wrote a CSV data-automation script that reduced manual data-processing effort by 60%.

EXPERIENCE

Rankbook Learning & Software, Mysuru — UI/UX Development Intern *Jun 2023 – Dec 2023*

- Redesigned interface flows and improved usability across the product, translating user feedback into iterative UI/UX changes.
- Collaborated with the development team to align design decisions with implementation constraints ahead of feature releases.

ACHIEVEMENTS

- Best Paper Presenter Award — International Conference

CERTIFICATIONS

- Python Workshop • UI/UX Development • Docker Essentials (in progress)

ADDITIONAL INFORMATION

Languages: English, Hindi, Kannada, Marwadi